

Intermediate Microeconomics — Week 2, Day 3

In-class Exercises: The Competitive Firm and Long-Run Entry

Perloff, Chs. 8–9 • work in pairs • solutions at the end

Exercise 1 — Cost curves, shutdown, and short-run supply [competency: firm-level cost analysis]

A price-taking firm has the total cost function

$$C(q) = 32 + 8q + 2q^2.$$

- (a) Write down marginal cost $MC(q)$, average variable cost $AVC(q)$, and average cost $AC(q)$.
 - (b) Find the shutdown price (minimum AVC) and the break-even price (minimum AC). At what output does AC reach its minimum?
 - (c) Derive the firm's short-run supply curve $q_s(P)$, stating the price range over which it produces.
 - (d) At $P = 16$ and again at $P = 40$: does the firm operate or shut down? Find its output and profit in each case.
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Exercise 2 — A demand boom, entry, and the long run [competency: market entry dynamics]

The market has many identical firms, each with the cost function from Exercise 1 (so $q_s = (P - 8)/4$ and $\min AC = 24$ at $q = 4$). Market demand is initially

$$Q_D = 200 - 5P.$$

- (a) The market is in long-run equilibrium. Find the long-run price, each firm's output, and the number of firms.
 - (b) A boom shifts demand to $Q'_D = 240 - 5P$. In the **short run** the number of firms is fixed at its part-(a) value. Find the new price, each firm's output, and each firm's profit. Confirm profit is positive.
 - (c) In the **long run**, how many firms operate, and how many entered?
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Reconvene: for Exercise 1 be ready to state the two threshold prices and *why* they differ; for Exercise 2, state what adjusts in the short run versus the long run.